

Sentiment Analysis using KNN - Practical Report

Aim

To perform sentiment analysis on a Twitter dataset using the K-Nearest Neighbors (KNN) algorithm.

Objectives

1. To understand text classification.
2. To preprocess Twitter text data.
3. To convert text into numerical format using TF-IDF.
4. To implement the KNN algorithm for sentiment classification.
5. To evaluate model performance using accuracy and confusion matrix.

Theory

Sentiment Analysis is a Natural Language Processing (NLP) technique used to identify emotions, opinions, or sentiments from textual data. Twitter datasets are commonly used for sentiment analysis because tweets contain real-world opinions. Machine Learning algorithms cannot understand raw text directly. Therefore, text must first be converted into numerical form using techniques such as TF-IDF (Term Frequency - Inverse Document Frequency). K-Nearest Neighbors (KNN) is a supervised machine learning algorithm that classifies data based on the nearest neighboring data points. In sentiment analysis, KNN compares a new tweet with existing tweets and predicts the sentiment based on similarity.

Important Formulas

1. TF (Term Frequency):
$$TF = (\text{Number of times a term appears in document}) / (\text{Total terms in document})$$
2. IDF (Inverse Document Frequency):
$$IDF = \log(\text{Total Documents} / \text{Documents containing the term})$$
3. TF-IDF Formula:
$$TF-IDF = TF \times IDF$$
4. Accuracy Formula:
$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions}) \times 100$$

Algorithm Steps

1. Import required libraries.
2. Load Twitter dataset.
3. Select text and sentiment columns.
4. Convert text into numerical vectors using TF-IDF.
5. Split dataset into training and testing sets.
6. Train KNN classifier.
7. Predict sentiments on test data.

8. Evaluate model using accuracy and confusion matrix.
9. Predict sentiment for custom tweet.

Code Explanation

- pandas → Used for dataset handling.
- matplotlib & seaborn → Used for visualization.
- TfidfVectorizer → Converts text into numerical features.
- train_test_split → Divides data into training and testing sets.
- KNeighborsClassifier → Implements KNN algorithm.
- accuracy_score → Calculates prediction accuracy.
- confusion_matrix → Evaluates classification performance.

Visualization Used

1. Confusion Matrix Heatmap
2. Sentiment Distribution Plot (optional)
3. Accuracy Plot (optional)

Expected Output

- Dataset successfully loaded.
- Text converted into vectors.
- KNN model trained successfully.
- Accuracy score displayed.
- Classification report displayed.
- Confusion matrix heatmap generated.
- Custom tweet sentiment predicted.

Advantages

1. Simple and easy to implement.
2. Effective for small datasets.
3. No complex training process required.
4. Good for classification problems.

Limitations

1. Slow for large datasets.
2. Requires proper value of K.
3. Sensitive to irrelevant features.
4. High memory usage.

Instructions to Run in Anaconda (.py File)

1. Open Anaconda Prompt.
2. Install required libraries using:

```
pip install pandas matplotlib seaborn scikit-learn
```

3. Open the folder containing the Python file.
4. Run the program using:

```
python filename.py
```

Instructions to Run in Jupyter Notebook (.ipynb File)

1. Open Anaconda Navigator.
2. Launch Jupyter Notebook.
3. Open the .ipynb file.
4. Upload dataset if needed.
5. Click 'Run All Cells'.
6. View outputs and graphs.

Important Viva Questions with Answers

Q1. What is Sentiment Analysis?

Ans: It identifies positive, negative, or neutral emotions from text data.

Q2. What is KNN?

Ans: KNN is a supervised machine learning algorithm that classifies data based on nearest neighbors.

Q3. Why TF-IDF is used?

Ans: TF-IDF converts text into numerical form for machine learning algorithms.

Q4. What is the role of train_test_split?

Ans: It divides the dataset into training and testing sets.

Q5. What is a confusion matrix?

Ans: It is a table used to evaluate classification performance.

Q6. What is accuracy?

Ans: Accuracy measures how many predictions are correct.

Q7. Why text preprocessing is important?

Ans: It removes unnecessary words and improves model performance.

Q8. What are neighbors in KNN?

Ans: They are the closest data points used for prediction.

Outcome

Successfully implemented sentiment analysis on Twitter data using the KNN algorithm with TF-IDF vectorization and evaluated the model using accuracy and confusion matrix.

Conclusion

The practical demonstrates how machine learning and NLP techniques can be used for sentiment analysis. KNN successfully classifies Twitter sentiments after converting text into numerical vectors using TF-IDF. The model performance can be evaluated using accuracy and confusion matrix visualization.